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PET361 MILESTONE2

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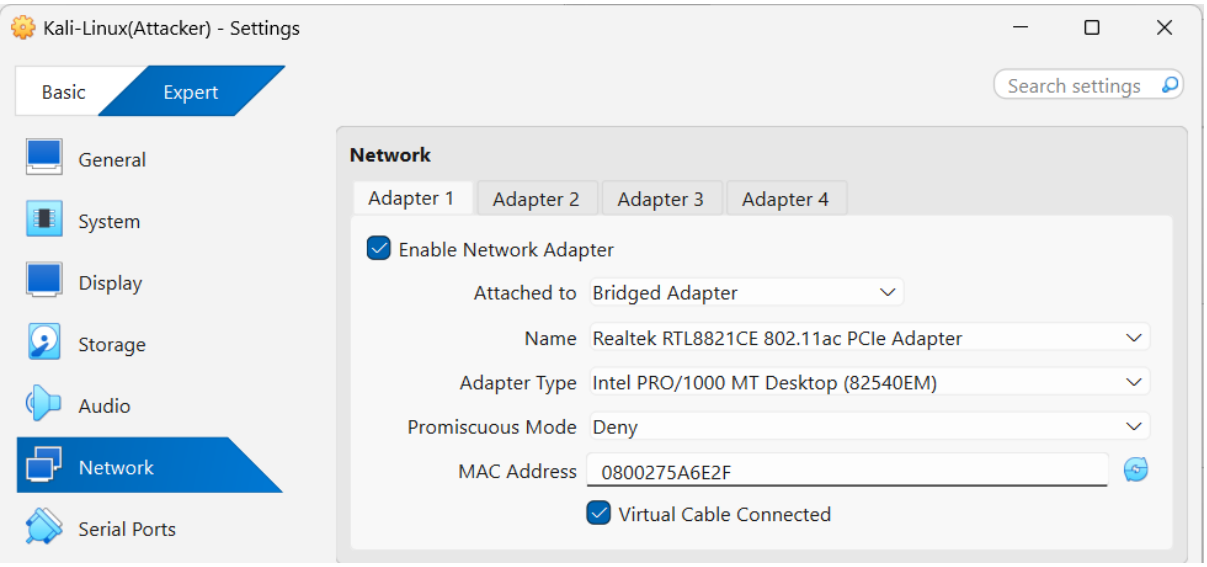
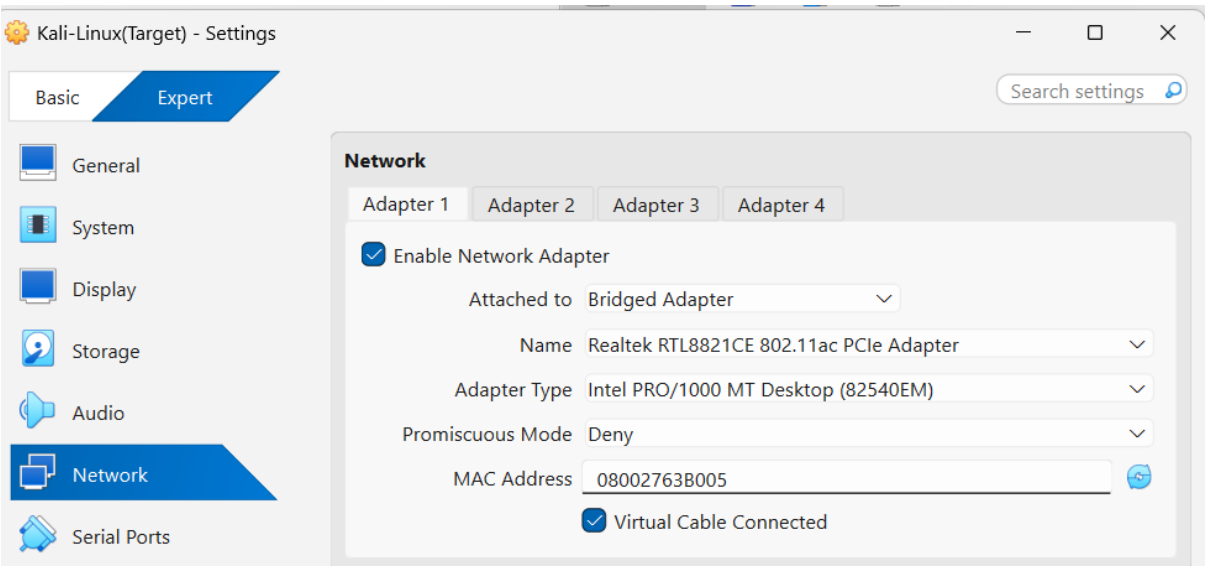
Introduction

This assignment demonstrates four Nmap scan types against a target Kali Linux. The scans performed are Stealth, TCP Connect, UDP and ACK. This report documents the results of each scan, explaining their methodology and compares their differences.

VirtualBox Networking Settings





I didn't use 'Internal Network' because I couldn't get an IP address so instead, I used 'Bridged Adapter' because both my attacker virtual machine and target virtual machine will get real Ip addresses. Both virtual machines can talk to each other without assigning IPs manually. I didn't use NAT because it hides the virtual machine behind the host and blocks direct VM-to-VM communication

IP Address

```
kali@kali: ~
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.8.133  netmask 255.255.255.0  broadcast 192.168.8.255
        inet6 fd8c:5ac1:3b2:8000:ff5d:b015:185b:5ad6  prefixlen 64  scopeid 0
x0<global>
        inet6 fe80::990b:803:1078:d344  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:63:b0:05  txqueuelen 1000  (Ethernet)
        RX packets 375  bytes 83614 (81.6 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 44  bytes 6002 (5.8 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 8  bytes 480 (480.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 8  bytes 480 (480.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

┌──(kali㉿kali)-[~]
└─$
```

```
kali@kali: ~
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.8.134  netmask 255.255.255.0  broadcast 192.168.8.255
        inet6 fd8c:5ac1:3b2:8000:71d:ae5c:cd86:c20e  prefixlen 64  scopeid 0x
0<global>
        inet6 fe80::b1f8:e96a:f394:1937  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:5a:6e:2f  txqueuelen 1000  (Ethernet)
        RX packets 412  bytes 89318 (87.2 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 49  bytes 5891 (5.7 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 8  bytes 480 (480.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 8  bytes 480 (480.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

┌──(kali㉿kali)-[~]
└─$
```

The IP Address for the Target Machine is **192.168.8.133** and the IP Address for the Attacker Machine is **192.168.8.134**

Ping Test

```
┌──(kali㉿kali)-[~]
└─$ ping 192.168.8.133 -c 4
PING 192.168.8.133 (192.168.8.133) 56(84) bytes of data.
64 bytes from 192.168.8.133: icmp_seq=1 ttl=64 time=16.9 ms
64 bytes from 192.168.8.133: icmp_seq=2 ttl=64 time=6.15 ms
64 bytes from 192.168.8.133: icmp_seq=3 ttl=64 time=3.01 ms
64 bytes from 192.168.8.133: icmp_seq=4 ttl=64 time=2.05 ms

--- 192.168.8.133 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3053ms
rtt min/avg/max/mdev = 2.054/7.031/16.916/5.904 ms

┌──(kali㉿kali)-[~]
└─$
```

I used the command ‘ping 192.168.8.133 -c 4’ to test connectivity. This is evidence that the virtual machines are properly configured and that the Target Machine was reachable.

‘-c’ (count) – this command tells ping how many packets to send

4 – this sends exactly 4 packets

With ‘-c 4’, ping runs indefinitely until you press Ctrl+C to stop it.

Ports

Kali has few open ports because it is secure by default. I opened some services for more scan results

```
kali@kali: ~
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ sudo service ssh start

┌──(kali㉿kali)-[~]
└─$ sudo service apache2 start

┌──(kali㉿kali)-[~]
└─$ sudo service vsftpd start
Failed to start vsftpd.service: Unit vsftpd.service not found.

┌──(kali㉿kali)-[~]
└─$ sudo service ssh status
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: >
     Active: active (running) since Thu 2026-03-26 16:40:02 EDT; 15min ago
 Invocation: daee87e8caa0434bab9bb354fe3ae663
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 26311 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUC>
   Main PID: 26313 (sshd)
      Tasks: 1 (limit: 2115)
     Memory: 2M (peak: 2.8M)
        CPU: 178ms
     CGroup: /system.slice/ssh.service
             └─26313 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startup>

Mar 26 16:40:02 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell>
Mar 26 16:40:02 kali sshd[26313]: Server listening on 0.0.0.0 port 22.
```

SSH (Server Shell) – protocol for secure remote access to a computer. It securely connects to and control a computer remotely. All communication is encrypted

Apache – this is a web server that serves websites. It listens for web browser requests and sends back web pages

Having more open ports gives you more complete scan results and better comparison

The command ‘sudo service vsftpd start’ failed it because it is not installed. I have 2 other ports, so it is fine if I don’t have this one.

```
┌──(kali㉿kali)-[~]
└─$ sudo ss -tulpn | grep LISTEN
[sudo] password for kali:
tcp   LISTEN 0      128                                0.0.0.0:22        0.0.0.0:
*    users:(("sshd",pid=26313,fd=6))


tcp   LISTEN 0      511                                      *:80              *:
*    users:(("apache2",pid=29696,fd=4),("apache2",pid=29695,fd=4),("apache2",
pid=29694,fd=4),("apache2",pid=29693,fd=4),("apache2",pid=29692,fd=4),("apach
e2",pid=29668,fd=4))
tcp   LISTEN 0      128                                   [::]:22           [::]:
*    users:(("sshd",pid=26313,fd=7))


┌──(kali㉿kali)-[~]
└─$ 
```

Scan Explanation and Results

Stealth Scan

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sS 192.168.8.133
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-27 09:10 -0400
Nmap scan report for 192.168.8.133
Host is up (0.0028s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE
22/tcp open  ssh
MAC Address: 08:00:27:63:B0:05 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.59 seconds
```

sudo nmap -sS 192.168.8.133

A stealth scan sends TCP SYN packets to target ports without completing the full three-way handshake. When a port is open, the target responds with SYN/ACK and the scanner sends an RST to abort the connection before it completes. This makes the scan less likely to be logged by applications

<u>Results:</u>

- Port 22 (SSH) was identified as open

- 999 ports were found to be closed returning “reset” (RST) response
- The scan completed in 1.59 seconds
- Sudo privileges were required to craft SYN packets

TCP Connect Scan

```
┌──(kali㉿kali)-[~]
└─$ nmap -sT 192.168.8.133
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-27 09:21 -0400
Nmap scan report for 192.168.8.133
Host is up (0.032s latency).
Not shown: 999 closed tcp ports (conn-refused)
PORT   STATE SERVICE
22/tcp open  ssh
MAC Address: 08:00:27:63:B0:05 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.90 seconds

┌──(kali㉿kali)-[~]
└─$
```

Nmap -st 192.168.8.133

TCP Connect Scan uses the operating systems connect () system call to complete the full TCP three-way handshake. For port 22 (SSH), the handshake was successfully completed, which confirms the service is listening. For closed ports, the target responded with an RST packet (indicated by “conn-refused”), which immediately terminates the connection attempt. It does not require root privileges, but it is more detectable because it completes the full connections that may be logged by the target system

<u>Result:</u>

- Port 22 (SSH) is open
- 999 TCP ports are closed
- Target runs as VirtualBox VM
- The scan completed in 1.90 seconds

UDP Scan

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sU 192.168.8.133 --top-ports 100
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-27 09:40 -0400
Nmap scan report for 192.168.8.133
Host is up (0.0026s latency).
All 100 scanned ports on 192.168.8.133 are in ignored states.
Not shown: 57 closed udp ports (port-unreach), 43 open|filtered udp ports (no
-response)
MAC Address: 08:00:27:63:B0:05 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 55.00 seconds
```

sudo nmap -sU 192.168.8.133 --top-ports 100

'--top -ports' – tells Nmap to scan the most commonly used ports

'100' – scan the top 100 most common ports

The target responds with an ICMP "Port Unreachable" message for closed ports. For open ports, the service may not respond which results in "open | filtered" state. There's no response for filtered ports, which is similar to open ports. The "open | filtered" state means Nmap can't determine whether the port is open or being filtered by a firewall. This is common with UDP scans because a lot of UDP services don't send responses to probes

<u>Results:</u>

- 57 UDP ports were confirmed as closed, returning "port-unreach' ICMP messages
- 43 UDP ports were marked as "open | filtered" because they didn't respond
- The scan completed in 55 seconds

ACK Scan

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sA 192.168.8.133
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-27 10:00 -0400
Nmap scan report for 192.168.8.133
Host is up (0.0027s latency).
All 1000 scanned ports on 192.168.8.133 are in ignored states.
Not shown: 1000 unfiltered tcp ports (reset)
MAC Address: 08:00:27:63:B0:05 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.65 seconds

┌──(kali㉿kali)-[~]
└─$
```

sudo nmap -sA 192.168.8.133

ACK Scan sends TCP packets with only the ACK flag set (no SYN flag). This scan doesn't determine if a port is open or closed, instead, it is used to map firewall rules by analysing how the target responds. The target responds with an RST (reset) packet for unfiltered ports which indicate that no stateful firewall is filtering ACK packets to the port. With filtered ports, there's no response or ICMP which indicates a firewall is dropping or is blocking the packets

Results:

- All 1000 scanned ports showed as "unfiltered"
- Each port returned an RST (reset) packet response
- The scan completed in 1.65 seconds

Comparison of scans

Stealth scan vs TCP Connect scan

- Both these scans identified port 22 as open
- The stealth scan requires root privileges, and it uses half-open connections
- TCP Connect scan does not require any root privileges and it uses full handshakes
- The results are identical when there's no firewall present

UDP scan vs TCP scan

- UDP scan produces no confirmed open ports
- UDP scan is much slower and have unclear results
- TCP scans are faster and have clear results

ACK scan

- This scan does not identify any open ports
- The scan identifes firewall filtering status
- There was no stateful firewall because all the ports are showed as "unfiltered"

Similarities

- SSH 22 is ran in the target system
- Noo UDP services were confirmed as open
- All the scans were done quickly except for UDP which required me to put a port limitation

References

Port Scanning Techniques. (n.d.). Retrieved from Nmap Project.: https://nmap.org/book/man-port-scanning-techniques.html

TCP ACK Scan (-sA). (n.d.). Retrieved from https://nmap.org/

TCP SYN (Stealth) Scan (-sS). (n.d.). Retrieved from Nmap Project. : https://nmap.org/book/synscan.html